

6.3.1 Magnetic fields

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) magnetic fields are due to moving charges or permanent magnets	
(b) magnetic field lines to map magnetic fields	
(c) magnetic field patterns for a long straight current-carrying conductor, a flat coil and a long solenoid	
(d) Fleming's left-hand rule	
(e) (i) force on a current-carrying conductor; $F = BIL\sin\theta$	
(ii) techniques and procedures used to determine the uniform magnetic flux density between the poles of a magnet using a current-carrying wire and digital balance	
(f) magnetic flux density; the unit tesla.	